

Exploratory Data Analysis (EDA) Report

Financial Health, Product Profitability & Country–Discount Analysis

1. Executive Summary

This EDA report analyzes the financial performance of products, monthly profitability, customer segments, countries, sales, discounts, COGS, units sold and profit margins. The analysis is presented through three Power BI dashboard pages:

1. **Executive Overview**
2. **Product and Profit Analysis**
3. **Country and Discount Insights**

The dashboards indicate that the business generated approximately **118.73M in total sales**, **101.83M in COGS**, and **16.89M in net profit**, resulting in an approximate **14% profit margin**. Among the products, **Paseo** is the strongest performer in terms of sales and profit, while **Carretera** has the lowest profit among the six products.

2. Objectives of the EDA

The main objectives of the analysis are:

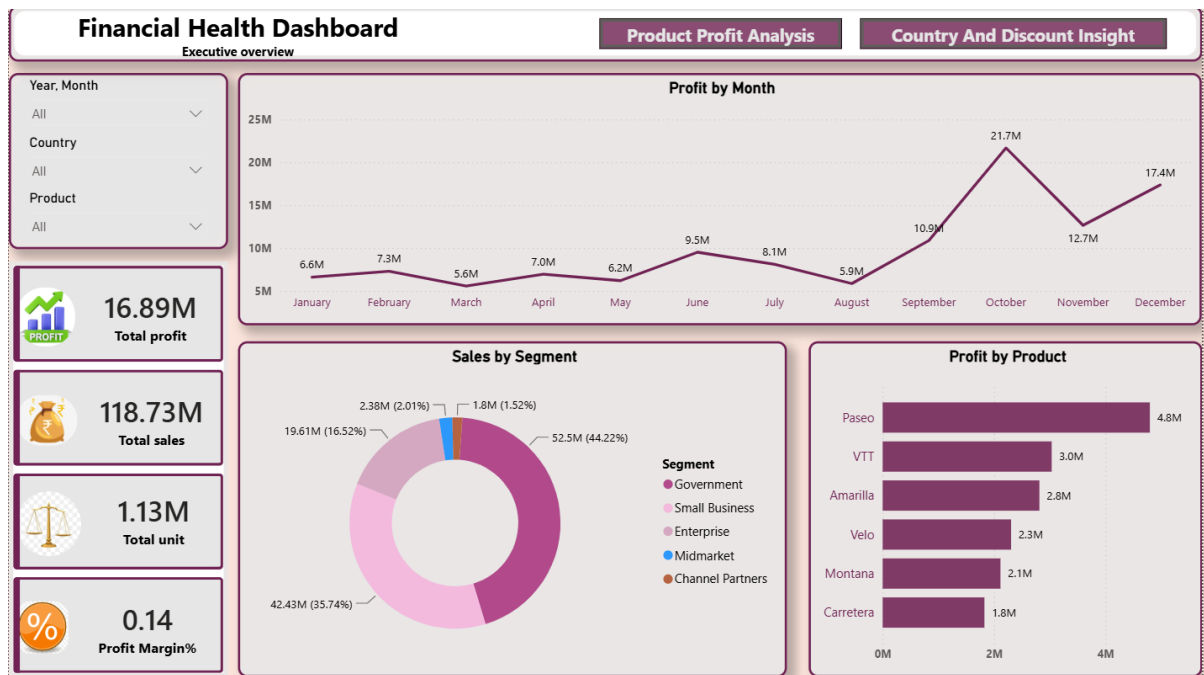
- Understand overall sales and profitability.
 - Identify the best and worst-performing products.
 - Analyze monthly profit trends.
 - Examine sales contribution by customer segment.
 - Compare product-level Sales, COGS and Profit.
 - Analyze country-wise sales and profit.
 - Understand the relationship between discounts and profitability.
 - Identify important business trends and opportunities.
 - Provide recommendations for improving financial performance.
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3. Key KPIs

KPI	Value	Interpretation
Total Sales	118.73M	Strong overall revenue generation
COGS	101.83M	Major portion of revenue is consumed by product costs
Net Profit	16.89M	Positive and healthy profitability
Profit Margin	14%	Indicates moderate profitability

KPI	Value	Interpretation
Total Units Sold	~1.13M	High sales volume
Number of Products	6	Product portfolio contains six major products
Countries	5	Sales are distributed across five countries
Total Discounts	9.21M	Discounts represent a significant business cost
Average Profit	24.13K	Average profit per transaction/record shown in the dashboard

4. Page 1 – Executive Overview



Monthly Profit Analysis

The monthly profit trend shows significant variation throughout the year.

Key observations:

- Profit starts at approximately **6.6M in January**.
- February increases to around **7.3M**.
- March declines to approximately **5.6M**.
- Profit rises again during April and May.
- June records approximately **9.5M**.
- July decreases to around **8.1M**.
- August is one of the weaker months at approximately **5.9M**.
- September shows a strong recovery to approximately **10.9M**.
- October is the highest-profit month at approximately 21.7M.**
- November decreases to around **12.7M**.
- December finishes strongly at approximately **17.4M**.

EDA Insight

October is the most profitable month, while August and March are relatively weak months. The large October increase should be investigated further to determine whether it was caused by higher sales volume, product mix, lower COGS, reduced discounts, or a seasonal/business event.

5. Sales by Customer Segment

The segment analysis shows that sales are concentrated in a few major customer segments.

Major observations:

- **Government:** approximately **52.5M**, representing around **44.22%** of sales.
- **Small Business:** approximately **42.43M**, representing around **35.74%**.
- **Enterprise:** approximately **19.61M**, representing around **16.52%**.
- Midmarket and Channel Partners contribute comparatively smaller amounts.

EDA Insight

Government and Small Business customers together contribute approximately **80% of total segment sales**. Therefore, these two segments are strategically important.

The business should focus on:

- Retaining high-value Government customers.
 - Expanding Small Business accounts.
 - Developing strategies to increase Enterprise and Midmarket contribution.
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6. Profit by Product

The product-level analysis clearly identifies the strongest products.

Product ranking by profit:

Rank	Product	Approx. Profit
1	Paseo	4.8M
2	VTT	3.0M
3	Amarilla	2.8M
4	Velo	2.3M
5	Montana	2.1M
6	Carretera	1.8M

EDA Insight

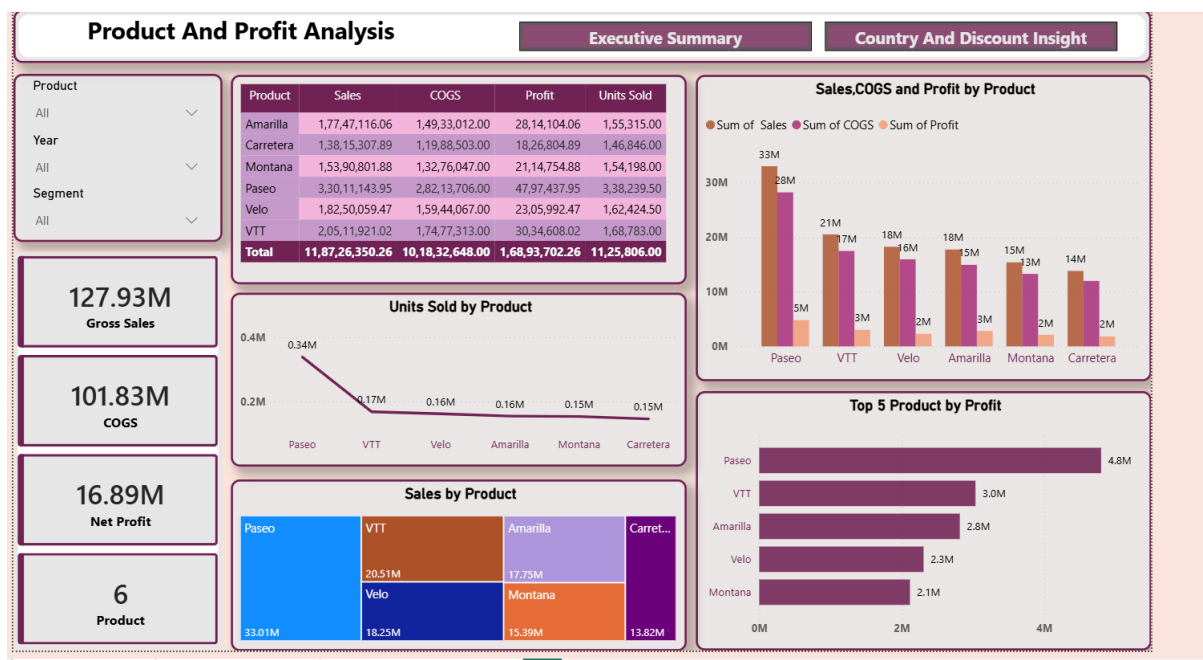
Paseo is the most profitable product, contributing approximately 4.8M profit. Carretera generates the lowest profit at approximately 1.8M.

Paseo should therefore be examined for its:

- Pricing strategy
- Sales volume
- COGS efficiency
- Customer demand
- Discount level

Carretera requires additional analysis to determine why its profitability is lower.

7. Page 2 – Product and Profit Analysis



Sales, COGS and Profit by Product

The product-level table provides a detailed comparison.

Major observations:

Paseo

- Sales: approximately **33.01M**
- COGS: approximately **28.21M**
- Profit: approximately **4.80M**
- Units Sold: approximately **338K**

VTT

- Sales: approximately **20.51M**
- COGS: approximately **17.48M**
- Profit: approximately **3.03M**
- Units Sold: approximately **169K**

Velo

- Sales: approximately **18.25M**
- COGS: approximately **15.94M**
- Profit: approximately **2.31M**
- Units Sold: approximately **162K**

Amarilla

- Sales: approximately **17.75M**
- COGS: approximately **14.93M**
- Profit: approximately **2.81M**
- Units Sold: approximately **155K**

Montana

- Sales: approximately **15.39M**
- COGS: approximately **13.28M**
- Profit: approximately **2.11M**
- Units Sold: approximately **154K**

Carretera

- Sales: approximately **13.82M**
- COGS: approximately **11.99M**
- Profit: approximately **1.83M**
- Units Sold: approximately **147K**

EDA Insight

Paseo dominates both **sales volume and profitability**. It generates approximately one-quarter of the product-level sales shown in the dashboard.

Although Carretera has the lowest sales, its profit is still positive. This suggests that the product is commercially viable but may require improvement in pricing, sales volume or cost efficiency.

8. Units Sold by Product

Paseo has the highest number of units sold at approximately **338K units**.

The remaining products are substantially lower:

- VTT: ~169K
- Velo: ~162K
- Amarilla: ~155K
- Montana: ~154K
- Carretera: ~147K

EDA Insight

There is a strong relationship between sales volume and overall sales performance. Paseo's significantly higher unit volume is an important reason for its leading revenue and profit position.

However, units alone should not be used to judge product profitability. Average selling price and cost per unit should also be analyzed.

9. Sales by Product

The treemap confirms that:

Paseo > VTT > Velo > Amarilla > Montana > Carretera

in terms of sales contribution.

Business interpretation

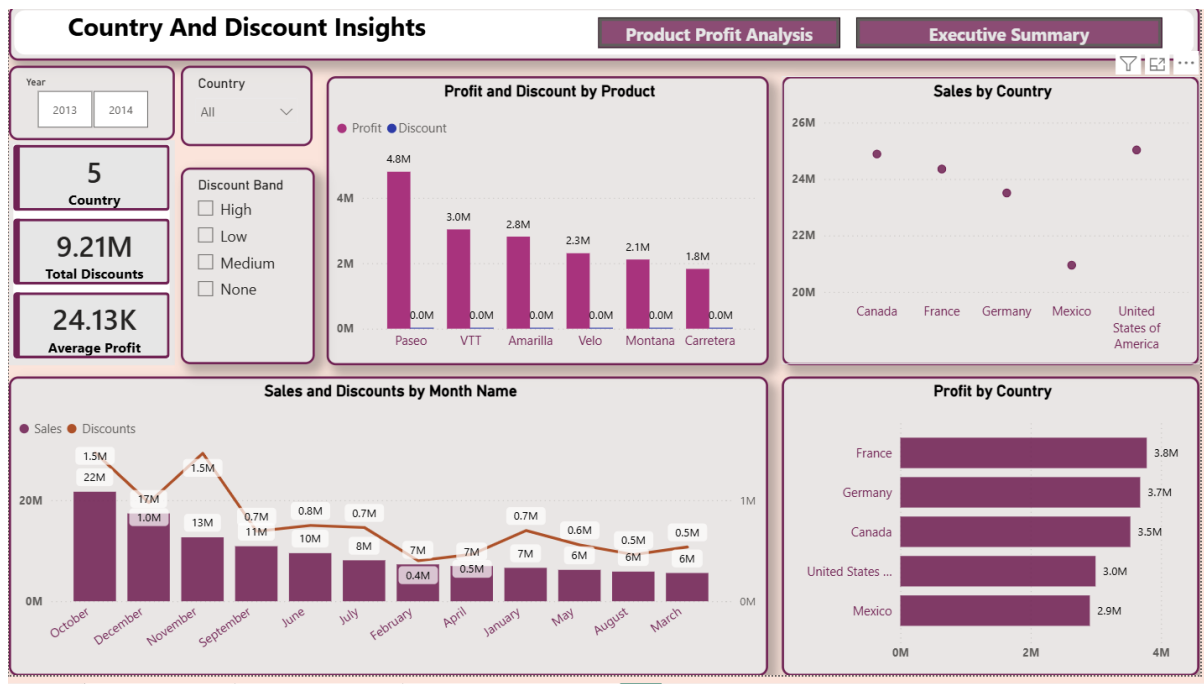
Paseo should be treated as a **core/high-performing product**.

VTT and Velo form the next important product group and could be used for cross-selling and portfolio expansion.

Carretera has the smallest sales contribution and should be reviewed for:

- Demand
 - Pricing
 - Discounting
 - Distribution
 - Product positioning
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10. Page 3 – Country and Discount Insights



Sales by Country

The dashboard covers five countries:

- Canada
- France
- Germany
- Mexico
- United States

Sales are approximately:

Country	Approx. Sales
United States	25.0M
Canada	24.9M
France	24.4M
Germany	23.5M
Mexico	21.0M

EDA Insight

The United States has the highest sales, while Mexico has the lowest.

However, the difference between countries is not extremely large, suggesting that the business has a relatively diversified geographical revenue base.

11. Profit by Country

The profit ranking is:

Rank	Country	Approx. Profit
1	France	3.8M
2	Germany	3.7M
3	Canada	3.5M
4	United States	3.0M
5	Mexico	2.9M

EDA Insight

An important finding is that the country with the highest sales is not necessarily the country with the highest profit.

France generates the highest profit despite the United States having slightly higher sales.

This indicates that profitability depends on more than revenue and may be influenced by:

- COGS
- Discounting
- Product mix
- Pricing
- Sales volume

12. Discount Analysis

The dashboard shows approximately **9.21M in total discounts**.

Discount bands include:

- High
- Medium
- Low
- None

EDA Insight

Discounts should be analyzed against profit rather than sales alone.

High discounts may increase sales volume but can reduce margins. Therefore, the company should identify whether discounted products actually generate incremental profit.

A useful future analysis would compare:

Discount % → Sales → COGS → Profit Margin

This would help identify the optimum discount strategy.

13. Sales and Discounts by Month

The monthly analysis indicates considerable variation in sales and discount levels.

Some months show relatively high sales along with higher discounts, while other months generate lower sales with lower discounts.

Important observation

October and November appear among the stronger sales periods, while several other months show considerably lower sales.

The relationship between sales and discounts should be investigated to answer:

Does increasing discounts actually generate enough additional sales to compensate for the reduction in margin?

This is an important business question for pricing strategy.

14. Product Profit and Discount Analysis

Paseo again ranks first with approximately **4.8M profit**, followed by:

- VTT – 3.0M
- Amarilla – 2.8M
- Velo – 2.3M
- Montana – 2.1M
- Carretera – 1.8M

The discount values shown in the visualization are comparatively small at the displayed product-level scale.

EDA Insight

The product profitability ranking remains consistent even when discount information is included. Paseo's strong profitability indicates that its performance is not simply dependent on discounting.

Final EDA Summary

High-performing Product: Paseo
Highest Sales Segment: Government
Highest Profit Country: France
Highest Profit Month: October
Total Sales: 118.73M
Total COGS: 101.83M
Total Profit: 16.89M
Profit Margin: ~14%
Total Discounts: 9.21M